EV Technology Course — Module 7: Vehicle Thermal Management and HVAC

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An EV has no engine throwing off waste heat, so every watt used to heat or cool the cabin comes from the battery, and in Indian city traffic air-conditioning can cut range by a third. This module covers the vehicle-level thermal system: separate coolant loops for battery and e-drive, the refrigerant circuit, heat pumps, integrated thermal modules, and how thermal strategy shapes range. (Battery-internal thermal design is in battery course Module 8.)

1. Learning objectives

- Identify the heat sources, heat sinks and target temperatures in an EV.
- Explain the refrigerant cycle and how a heat pump heats the cabin efficiently.
- Explain chillers, multi-way valves and integrated thermal modules.
- Quantify the range impact of heating and air-conditioning.
- Describe thermal strategy: preconditioning, waste-heat recovery, cabin comfort vs range.

2. Fundamentals

Three things need temperature control, each with a different target:

Item	Target	Problem when wrong
Battery	~20–40 °C	Cold: low power, slow charging; hot: fast ageing
Motor + power electronics	Coolant up to ~60–70 °C	Overheating derates power; too cold is fine
Cabin	~22–25 °C for people	Comfort, demisting for safety

A petrol car heats its cabin almost for free with engine waste heat. An EV's motor and inverter produce far less waste heat (they are 90%+ efficient), so heating must come from a **PTC heater** (electric resistance, efficiency ~100%) or a **heat pump** (moves heat, 200–300% effective). Cooling in both cases uses an electrically driven AC compressor.

3. Detailed technical explanation

The refrigerant cycle (AC). An electric compressor squeezes refrigerant gas, making it hot; it releases heat in a condenser; it passes through an expansion valve, becoming cold; it absorbs heat in an evaporator (cooling the cabin air) and returns to the compressor.

Heat pump. The same cycle run the other way: valves reconfigure the circuit so the cold side absorbs heat from outside air (or from battery/e-drive coolant) and the hot side heats the cabin. Coefficient of performance (COP) is typically 2–3 at mild temperatures, dropping as it gets colder.

Chiller. A plate heat exchanger where refrigerant cools the battery coolant loop: essential for fast charging and Indian summers.

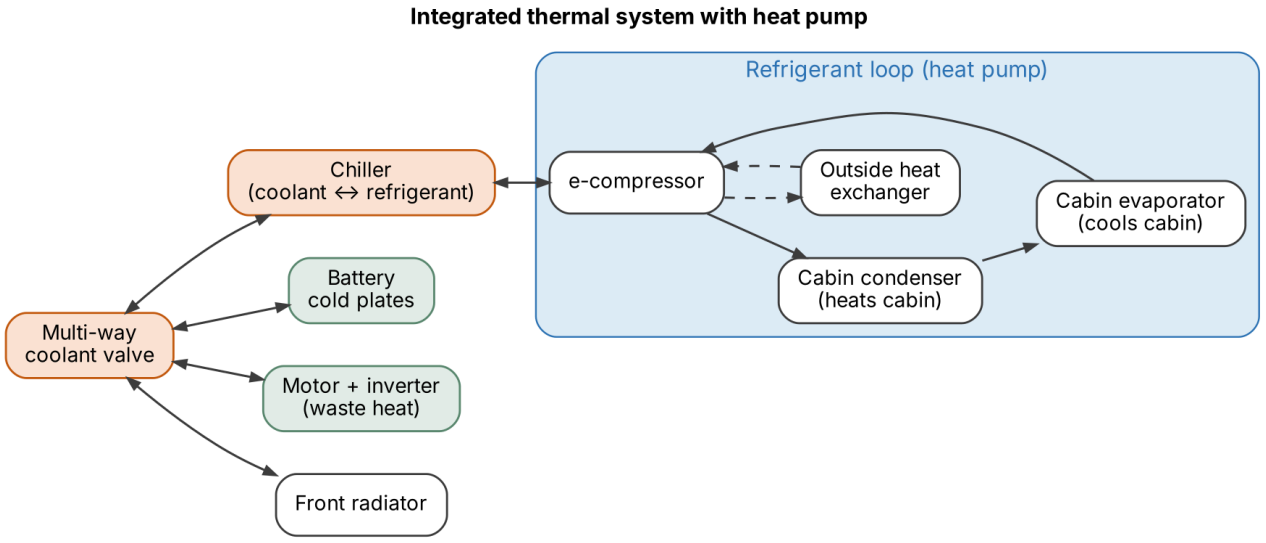
Multi-way valves and integrated thermal modules (ITM). Instead of separate pumps, valves and pipes, one module contains pumps, valves, chiller and sensors, and a multi-way valve routes coolant between battery, e-drive, radiator and chiller depending on mode (e.g. Tesla's "Octovalve"). Benefits: fewer parts and joints, waste-heat recovery, less refrigerant.

Refrigerants. R1234yf is common today; R744 (CO_2) works well for heat pumps in cold climates but needs very high pressures; R134a is being phased out in many markets.

Thermal modes (examples)

Mode	What happens
Summer driving	Compressor cools cabin; chiller cools battery if needed; radiator cools e-drive
Fast charging in heat	Chiller works hard on battery; cabin cooling may be reduced
Winter driving	Heat pump heats cabin using outside air and e-drive waste heat
Preconditioning	Battery warmed (or cooled) before arriving at a fast charger
Parked in sun	Cabin overheat protection; battery cooling if needed

4. Diagram: integrated thermal system with heat pump



5. Step-by-step: a hot Gurgaon afternoon, then a fast charge

1. Ambient 42 °C; the driver sets 23 °C. The compressor runs at high speed, drawing 2–3 kW to pull the cabin down; then about 1–2 kW to hold it.
2. Battery temperature rises during driving; the chiller diverts some refrigerant capacity to keep the pack below about 35 °C.
3. Navigation shows a DC charger ahead; the thermal manager pre-cools the battery so charging can start at full power.
4. During charging, battery heat rises sharply; the chiller takes priority; the cabin stays at a comfortable but slightly higher setpoint.
5. After charging, the system returns to cabin priority.

6. Engineering equations

Heat carried by a fluid:

```
\dot{Q} = \dot{m} c_p \Delta T
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Coefficient of performance:

```
COP = \frac{\dot{Q}_{delivered}}{P_{electric}}
```

Extra energy per km from HVAC:

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\Delta c\ (\text{Wh/km}) = \frac{P_{HVAC}\ (\text{W})}{v\ (\text{km/h})}
```

The last equation shows why HVAC hurts most in slow traffic: the same kilowatts are spread over fewer kilometres.

7. Worked examples

Example 1: PTC vs heat pump in winter. Cabin needs 4 kW of heat at 40 km/h. PTC (COP 1): 4,000 ÷ 40 = **100 Wh/km** extra. Heat pump (COP 2.5): 1.6 kW → **40 Wh/km** extra.

Example 2: Indian summer city range. Base consumption 120 Wh/km; AC 2 kW at an average 30 km/h adds 67 Wh/km. Range from 58.4 kWh usable: without AC **487 km**, with AC **313 km**, a 36% drop.

Example 3: e-drive coolant flow. Motor + inverter losses 3 kW, coolant rise 5 K, c_p 3,500 J/(kg·K): $\dot{m}$ = 0.171 kg/s ≈ **9.7 L/min**.

8. Real-world examples

- **Tesla Model Y:** heat pump with Octovalve; uses motor inefficiency deliberately to heat the battery when needed.
- **Hyundai E-GMP:** heat pump recovering waste heat from the e-drive.
- **Indian EVs:** most have liquid-cooled batteries linked to the AC system; heat pumps are less common because heating demand is small in most of India.
- **Electric buses:** large roof-mounted AC units; HVAC can be a major share of consumption in hot cities.

9. Manufacturing applications

- **More circuits to fill:** separate coolant loops and a refrigerant loop need vacuum filling and leak testing at the line.
- **Integrated thermal modules** arrive pre-tested from suppliers, cutting plant steps and leak points.
- **Refrigerant handling:** charge quantity accuracy, leak detection (helium or sniffer), environmental rules.
- **Costing:** heat pump adds cost (valves, extra heat exchangers, controls); justify by range gain in cold markets; in India, cooling capacity and chiller sizing matter more.

10. Common mistakes and misconceptions

- **"Heat pumps always save energy."** Benefit shrinks in extreme cold and is small where heating is rarely needed.

- **"AC barely affects EV range."** In slow, hot traffic it can be one of the biggest loads.
- **"Motors and inverters need the same cooling as the battery."** They tolerate much higher coolant temperatures.
- **"Thermal management is only about the battery."** Cabin comfort and demisting are safety and customer-satisfaction requirements.

11. Practical case study: summer range complaints

Problem. Owners in Delhi and Hyderabad report summer city range 35–40% below the claimed figure.

Analysis. High AC load at low average speeds, sun-soaked cabins needing long pull-down, battery cooling running while parked.

Actions. Remote pre-cooling while plugged in (using grid energy), better glass and insulation, smarter recirculation, efficient variable-speed compressor control, and honest real-world range guidance.

Lesson. In hot markets, HVAC efficiency is a range feature.

12. Revision notes

- Three targets: battery ~20–40 °C, e-drive up to ~65 °C coolant, cabin ~23 °C.
- Heat pump COP 2–3 vs PTC 1.
- Chiller links refrigerant to battery coolant.
- Integrated modules and multi-way valves cut parts and recover waste heat.
- HVAC Wh/km = HVAC W ÷ speed (km/h).

13. Glossary

Term	Meaning
PTC heater	Self-regulating resistive heater
Heat pump	Refrigerant cycle used to move heat into the cabin
COP	Coefficient of performance
Chiller	Refrigerant-to-coolant heat exchanger
ITM	Integrated thermal module
R1234yf / R744	Common refrigerant / CO_2 refrigerant

Term	Meaning
Preconditioning	Heating or cooling before driving or charging

14. Knowledge check

Conceptual

1. Why do EVs need dedicated cabin heating?
2. How does a heat pump deliver more heat than the electricity it uses?
3. Why does HVAC hurt range more in slow traffic?
4. Why are battery and e-drive coolant loops kept at different temperatures?
5. When is a heat pump not worth its cost?

Numerical

1. Heat pump COP 3 delivering 3 kW of heat: electrical power?
2. AC load 1.5 kW at an average 60 km/h: extra Wh/km?
3. Battery and e-drive reject 5 kW through a chiller with system COP 2.5: compressor power?

Application

1. Specify a thermal concept for an Indian compact EV that must fast-charge in 45 °C heat.
2. List the thermal-related end-of-line tests.

15. Answer key

Conceptual

1. The motor and inverter are so efficient that their waste heat is too small and too cool to warm the cabin reliably.
2. It moves existing heat from outside air or waste heat into the cabin; the electricity drives the compressor rather than creating heat directly.
3. HVAC draws constant power per hour; at low speed, each hour covers fewer kilometres, so Wh/km rises.
4. Batteries age fast above ~35–40 °C; motors and inverters run fine with much warmer coolant, which also makes it easier to reject heat to the air.
5. In hot climates with little heating demand, or very low-cost models where the extra hardware cost outweighs the small range gain.

Numerical

1. 3 ÷ 3 = **1 kW**.
2. 1,500 ÷ 60 = **25 Wh/km**.
3. 5 ÷ 2.5 = **2 kW**.

Application

1. Liquid-cooled battery with a chiller sized for fast-charge heat at 45 °C ambient; variable-speed electric compressor; navigation-based pre-cooling; cabin pre-cool while plugged in; e-drive on a separate radiator loop; PTC heater only (heat pump optional); good glass and insulation.
2. Coolant and refrigerant vacuum-fill and leak test, refrigerant charge weight, pump and valve function, compressor run test, sensor plausibility, HVAC outlet temperature check, chiller function during a short charge test.

16. Further learning

- Hendricks et al. and NREL publications on EV thermal management and HVAC energy use.
- SAE J2843 (R1234yf systems); SAE papers on EV heat pumps.
- Tesla and Hyundai technical presentations on integrated thermal systems.

Next: Module 8, chassis, body, crash safety and lightweighting.